

PC-08 · PRESERVATIVES & CHELANTS · PERSONAL CARE

HS GXY

Benzisothiazolinone & Propanediol

INCI	CAS	EC	FAMILY	SUPPLIED AS
Benzisothiazolinone & Propanediol	2634-33-5	—	PC-08	Liquid

01 POSITIONING

The preservative decision is a market-access decision — we supply the globally accepted systems.

02 HOW IT WORKS

- 01

Membrane disruption

Phenoxyethanol permeabilizes microbial membranes, halting proliferation across a broad range.
- 02

Booster synergy

Ethylhexylglycerin lowers interfacial tension on microbes, cutting the phenoxyethanol level needed.
- 03

Broad + green options

Benzisothiazolinone blends (GXY-type) cover rinse-off and industrial where cost governs.

03 APPLICATIONS

- Creams, lotions and serums
- Shampoos and rinse-off
- Wet wipes
- Household and I&I (BIT blends)
- Water-based industrial fluids

04 FORMULATION GUIDE

USE LEVEL	0.3–1%
PH & CONDITIONS	Phenoxyethanol effective pH 3–10; BIT systems favor pH > 5

- Add in the cool-down below 40 °C; run a challenge test on the finished formula, not the raw material.
- Respect regional leave-on limits (phenoxyethanol commonly ≤ 1%).

05 TYPICAL SPECIFICATION

Appearance	Clear liquid
Assay / active	per grade
pH	per TDS
Global compliance	EU Annex V / regional

Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE — DIRECTIONAL

Challenge-test pass margin (class data)
Phenoxyethanol + booster



Global label acceptance
Phenoxyethanol systems



Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.